



DOW™ LDPE 722

The Dow Chemical Company - Low Density Polyethylene Resin

Wednesday, November 30, 2016

General Information

Product Description

Dow™ LDPE 722 is used in flexible packaging and paperboard coating applications such as liquid/juice, laminate tube, condiment pouches, dry foods packaging, snack foods packaging, moist foods packaging, sugar pouches, lidding stock and medical packaging. DOW LDPE extrusion coating resins provide optimal neck-in and draw-down performance with minimal taste/odor contribution.

DOW Polyethylene 722 is a broad molecular weight distribution homopolymer designed to offer good impact strength and crack resistance, with excellent flexibility. The resin has good processability over a wide range of molding conditions.

- Typical applications include caps/closures
- Good impact, ESCR with excellent flexibility

Complies with:

- CANADIAN HPFB NO OBJECTION (WITH LIMITATIONS)
- EU, No 10/2011
- U.S. FDA 21 CFR 177.1520 (c) 2.2
- U.S. FDA DMF

Consult the regulations for complete details.

General

Material Status	• Commercial: Active		
Availability	• Asia Pacific	• Latin America	• North America
Agency Ratings	• DMF Unspecified Rating • EU No 10/2011	• FDA 21 CFR 177.1520(c) 2.2 • HPFB (Canada) No Objection ¹	
Forms	• Pellets		
Processing Method	• Extrusion Coating	• Injection Molding	

ASTM & ISO Properties ²

Physical	Nominal Value	Unit	Test Method
Specific Gravity	0.920		ASTM D792
Melt Mass-Flow Rate (190°C/2.16 kg)	8.0	g/10 min	ASTM D1238
Environmental Stress-Cracking Resistance ³ 122°F, 100% Igepal, F50	< 1.00	hr	ASTM D1693
Mechanical	Nominal Value	Unit	Test Method
Tensile Strength ³ (Yield)	1200	psi	ASTM D638
Tensile Strength ³ (Break)	1400	psi	ASTM D638
Tensile Elongation ³ (Yield)	4.0	%	ASTM D638
Tensile Elongation ³ (Break)	500	%	ASTM D638
Flexural Modulus - 2% Secant ³	34000	psi	ASTM D790B
Coefficient of Friction	0.60		ASTM D1894
Films	Nominal Value	Unit	Test Method
Seal Initiation Temperature ⁴	221	°F	Internal Method
Water Vapor Transmission Rate	1.7	g·mil/100in ² /atm/24 hr	ASTM F1249
Impact	Nominal Value	Unit	Test Method
Tensile Impact Strength ^{5, 3}	130	ft·lb/in ²	ASTM D1822

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Hardness	Nominal Value	Unit	Test Method
Durometer Hardness ³ (Shore D)	43		ASTM D2240
Thermal	Nominal Value	Unit	Test Method
Deflection Temperature Under Load ³ (66 psi, Unannealed)	99.0	°F	ASTM D648
Brittleness Temperature ³	-76.0	°F	ASTM D746
Vicat Softening Temperature	190	°F	ASTM D1525
Melting Temperature (DSC)	224	°F	Internal Method
Peak Crystallization Temperature (DSC)	204	°F	Internal Method
Additional Information	Nominal Value	Unit	Test Method
Melt Temperature - Recommended	600 to 630	°F	Internal Method

Fabrication Conditions For Extrusion Coating Film:

- Screw Size: 3.5 in. (89 mm); 30:1 L/D
- Screw Type: Single Flight with Maddock Mixer
- Die Gap: 20 mil (0.508 mm)
- Melt Temperature: 625°F (329°C)
- Output: 250 lb/hr
- Screw Speed: 90 rpm

Processing Information

Extrusion	Nominal Value	Unit	Test Method
Maximum Line Speed	25.0	ft/sec	Internal Method
Minimum Coating Thickness	0.30	mil	Internal Method
Minimum Coating Weight	4.4	lb/ream	Internal Method
Neck-in (610°F, 1.0 mil)	2.0	in	Internal Method

Notes

¹ With limitations

² Typical properties: these are not to be construed as specifications.

³ Molded and tested in accordance with ASTM D4976.

⁴ Temperature at which 1 lb/in (4.4 N/25.4 mm) heat seal strength is achieved.

Heat Seal Strengths, Topware HT Tester 0.5 S dwell, 40 psi bar pressure, pull speed 250 mm/sec.

⁵ Type S

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